

Transfer Learning for Freshness Classification of Fruits and Vegetables: A Comparative Study of Backbone Architectures and Training Conditions

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Abstract—Automated freshness classification of fruits and vegetables is a critical task for food safety, supply chain management, and retail quality control. This paper presents a systematic study of transfer learning applied to this problem, evaluating four convolutional neural network backbone architectures — ResNet-18, MobileNetV3-Small, EfficientNet-B0, and EfficientNet-B2 — under four training conditions that vary the degree of backbone fine-tuning and the presence of a non-linear projection head. Experiments are conducted on six publicly available datasets comprising over 55,000 images across multiple fruit and vegetable categories, including binary (fresh/rotten) and three-class (fresh/formalin/rotten) annotation schemes. On the two datasets for which published baselines exist, our models exceed the state of the art: on the apple/banana/orange benchmark our best single model reaches a perfect 100% accuracy and a 16-model ensemble attains 99.96%, surpassing the previously reported 99.61%; on the peach/pomegranate/strawberry benchmark our ensemble reaches 96.68%, surpassing the published 95.0%. A key finding is that the optimal training condition differs by backbone family: fine-tuning the last residual block benefits ResNet-18 (C4), whereas MobileNetV3 and both EfficientNet variants perform best with a completely frozen backbone and projection head (C3). We further quantify domain shift on a held-out set of 219 real-world mobile-camera photos of two fruits, showing that a model's per-fruit accuracy is governed by whether that fruit appeared in training (banana F1 of 71% when seen versus 42% when unseen), and we demonstrate that a lightweight head-only fine-tuning step closes most of this gap, raising macro F1 from 86.3% to 94.5%.

Index Terms—transfer learning, freshness classification, convolutional neural networks, EfficientNet, MobileNet, ResNet, model ensemble, domain shift, food quality inspection, fine-tuning

I. INTRODUCTION

Food freshness is a critical quality attribute that directly affects consumer health, retail profitability, and supply chain efficiency. Manual inspection of fruits and vegetables is time-consuming, subjective, and difficult to scale to the volumes required by modern distribution systems. Automated computer vision systems offer a scalable and objective alternative, capable of processing large volumes of produce at low cost.

Recent advances in deep learning, particularly convolutional neural networks (CNNs) pre-trained on large-scale image datasets, have demonstrated strong performance on a wide range of visual classification tasks. Transfer learning [1] leverages

these pre-trained representations to build effective classifiers from limited domain-specific data — an important property for food quality applications where labeled datasets are expensive to collect.

Despite the growing body of work on food classification and freshness detection, a systematic comparison of backbone architectures and fine-tuning strategies under controlled conditions remains lacking. Furthermore, the domain shift between publicly available dataset images (typically acquired in controlled photographic conditions) and real-world images captured by mobile devices has received limited attention.

This paper addresses both gaps through the following contributions:

- A controlled comparison of four CNN backbone architectures (ResNet-18, MobileNetV3-Small, EfficientNet-B0, EfficientNet-B2) and four training conditions (C1–C4) across six benchmark datasets totalling 55,040 images, in both binary and six-class (fruit $\times$ state) formulations.
- State-of-the-art results on the two datasets with published baselines: a perfect 100% accuracy (best single model) and a 99.96% 16-model ensemble on the apple/banana/orange benchmark, and a 96.68% ensemble on the peach/pomegranate/strawberry benchmark, both exceeding previously reported figures.
- The identification of a consistent interaction between backbone family and optimal training condition: C4 (partial fine-tuning + projection head) for ResNet-18 vs. C3 (frozen backbone + projection head) for the MobileNet and EfficientNet families.
- A per-fruit domain-shift analysis on 219 real mobile-camera photos showing that generalization tracks training exposure, together with a head-only fine-tuning recipe that raises in-domain macro F1 from 86.3% to 94.5%.
- An open-source pipeline implementing the full experimental workflow from data preparation to ensemble construction and benchmark reporting.

The remainder of the paper is organized as follows. Section II reviews related work. Section III describes the methodology.

Section IV presents experimental results. Section V discusses key findings. Section VI concludes the paper.

II. RELATED WORK

A. Food Quality Classification with Deep Learning

Deep learning has been widely applied to food recognition and quality assessment [2]. Several works specifically target freshness detection using CNNs trained from scratch or with transfer learning [3], [4]. Bangladeshi et al. evaluated multiple deep learning architectures for fresh/rotten detection of fruits and vegetables, finding that pre-trained models significantly outperform models trained from scratch [5].

B. Backbone Architectures

ResNet [6] introduced skip connections to address the vanishing gradient problem, enabling deeper networks that generalize well across domains. EfficientNet [7] proposed compound scaling — simultaneously adjusting network depth, width, and input resolution — achieving superior ImageNet accuracy with fewer parameters than comparably-performing ResNet variants. EfficientNet-B0 and B2 achieve 77.1% and 80.1% top-1 accuracy on ImageNet respectively, compared to 69.8% for ResNet-18, with approximately half the parameters.

C. Transfer Learning Strategies

The most common transfer learning strategies for image classification range from linear probing (training only the final classifier) to full fine-tuning (updating all weights) [1]. Kornblith et al. [8] showed that better ImageNet features transfer better across tasks. The intermediate strategy of fine-tuning only the later layers — which capture more task-specific features — is widely adopted as a balance between adaptation and catastrophic forgetting [9]. Combining multiple models into an ensemble by averaging their predicted class probabilities is a long-established technique for reducing variance and exploiting complementary errors across classifiers [10]; its benefit is largest when the constituent models are diverse and the task is not already saturated.

D. Prior Results on the Benchmark Datasets

Two of the datasets used here have published baselines that enable a direct comparison. On the Kaggle apple/banana/orange collection, Palakodati et al. [11] trained a custom CNN and reported 97.82% accuracy, while Chakraborty et al. [12] obtained 99.61% with a MobileNetV2 transfer-learning model. On the Mendeley peach/pomegranate/strawberry collection, Sharma and Kumar [13] fine-tuned ResNet50 and reported 95.0% accuracy. All three report accuracy as their primary metric, which we adopt for the comparison in Section IV.

III. METHODOLOGY

A. Datasets

Six publicly available datasets are used in this study (Table I). Together they cover 55,040 images across fruits, vegetables, and citrus varieties, with binary freshness labels (fresh/rotten) in five datasets and a three-class annotation (fresh/formalin/rotten)

in one. The FruitVision dataset is particularly relevant as the formalin class presents a significant challenge: chemically-treated samples are visually similar to fresh specimens, analogous to the difficulty of detecting subtle spoilage in fermented foods.

TABLE I: Datasets used in this study. [†]Three-class annotation.

Dataset	ID	Images	Classes	Source
FruitVision [†]	MFV	10,154	3	Mendeley
Fruits Classification	MFR	1,655	2	Mendeley
Lemon Varieties	MLM	1,956	2	Mendeley
Fruits Quality	KFQ	359	2	Kaggle
Fruits Fresh/Rotten	KFR	13,599	2	Kaggle
Fresh & Stale	KFS	27,317	2	Kaggle
Total		55,040		

B. Data Preprocessing

All images are resized so the shortest side is 256 pixels and then center-cropped to 224×224 for validation. Training images undergo the following augmentation pipeline: random resized crop to 224×224 , random horizontal flip, $\pm 30^\circ$ rotation, color jitter (brightness, contrast, saturation, hue), and Gaussian blur (kernel size 3). All images are normalized with ImageNet channel statistics ($\mu = [0.485, 0.456, 0.406]$, $\sigma = [0.229, 0.224, 0.225]$). A stratified 80/20 train/validation split with fixed random seed 42 is applied to all datasets.

C. Backbone Architectures

Four pre-trained CNN backbones are evaluated, all initialized with ImageNet weights from the `torchvision` library:

- **ResNet-18** [6]: 11.2M parameters, 512-dimensional feature output from global average pooling.
- **MobileNetV3-Small**: 2.5M parameters, 576-dimensional feature output; the most compact backbone, designed for mobile and embedded inference.
- **EfficientNet-B0** [7]: 5.3M parameters, 1280-dimensional feature output.
- **EfficientNet-B2** [7]: 9.1M parameters, 1408-dimensional feature output.

D. Training Conditions

Four training conditions form an ablation study over two independent dimensions: (1) whether the backbone is partially fine-tuned, and (2) whether a non-linear projection head is used before the classifier. Table II summarizes all four combinations.

TABLE II: Training conditions — ablation over fine-tuning and head architecture.

ID	Backbone	Head	Description
C1	Frozen	Linear only	Linear probe
C2	Layer4 free	Linear only	Partial fine-tuning
C3	Frozen	Proj. + Linear	Full head, frozen backbone
C4	Layer4 free	Proj. + Linear	Full head + fine-tuning

C1 — Linear probe (frozen backbone, linear head).

All backbone weights are locked throughout training. Only a

single linear layer $\mathbf{W} \in \mathbb{R}^{d_{\text{out}} \times C}$ is optimized, where d_{out} is the backbone output dimension and C the number of classes. This is the classical *linear probe* setting [8], which evaluates the quality of pre-trained features without any domain adaptation. It is the fastest and most data-efficient condition, but cannot correct for any mismatch between ImageNet features and the target domain.

C2 — Partial fine-tuning (layer4 free, linear head). The last residual block of the backbone (`layer4`, comprising approximately 2M parameters for ResNet-18 and proportionally more for EfficientNet) is unfrozen and trained jointly with the linear classifier. All earlier layers remain frozen. This allows the highest-level feature detectors to adapt toward freshness-specific representations while retaining the general low-level and mid-level features learned from ImageNet. The linear head is the same as C1, providing no additional non-linear capacity beyond the unfrozen backbone block.

C3 — Projection head with frozen backbone. The backbone is fully frozen (as in C1), but the simple linear classifier is replaced by a two-layer non-linear projection head followed by a linear output layer:

$$\mathbf{h} = \text{ReLU}(\mathbf{W}_2 \text{ReLU}(\mathbf{W}_1 \mathbf{f})), \quad \hat{y} = \mathbf{W}_3 \mathbf{h} \quad (1)$$

where $\mathbf{f} \in \mathbb{R}^{d_{\text{out}}}$ is the backbone feature vector, $\mathbf{W}_1 \in \mathbb{R}^{256 \times d_{\text{out}}}$, $\mathbf{W}_2 \in \mathbb{R}^{128 \times 256}$, and $\mathbf{W}_3 \in \mathbb{R}^{C \times 128}$. The non-linear projection can reshape the feature space to form more discriminative decision boundaries without requiring backbone fine-tuning. This condition is particularly relevant when the pre-trained features are already highly expressive, as is the case for larger EfficientNet variants.

C4 — Full head with partial fine-tuning. This condition combines the projection head of C3 with the backbone fine-tuning of C2. The last backbone block (`layer4`) and the full projection head are optimized jointly, using a dual learning rate strategy: $\text{lr} = 10^{-3}$ for the projection head and classifier, and $\text{lr}_{\text{backbone}} = 10^{-5}$ for the backbone block. The reduced backbone learning rate is essential to prevent catastrophic forgetting [9]: aggressive updates to the backbone destroy the general-purpose representations learned during ImageNet pre-training, effectively resetting the transfer learning benefit. By using a learning rate two orders of magnitude smaller, the backbone adapts *gradually* toward freshness-relevant features while preserving the hierarchical feature structure inherited from pre-training.

All models are trained for 60 epochs using the Adam optimizer ($\beta_1 = 0.9$, $\beta_2 = 0.999$) with cosine annealing learning rate schedule and weight decay $\lambda = 10^{-4}$. Batch size is 32.

E. Label Modes

Each dataset is trained under two label formulations. In **state** mode the classes are the freshness states only (fresh/rotten, or fresh/formalin/rotten), so the classifier is fruit-agnostic. In **fruit_state** mode each fruit $\times$ state combination is a distinct class (e.g. `apple_fresh`, `apple_rotten`), yielding six classes for the two three-fruit benchmark datasets. The

fruit_state formulation matches the protocol of the published baselines and is therefore used for the head-to-head comparison (Section IV-C).

F. Model Ensemble

For the two benchmark datasets, the sixteen fruit_state models (four backbones $\times$ four conditions) are combined into a single classifier by averaging their predicted class-probability vectors. Let $\mathbf{p}_i^{(m)} \in \Delta^{C-1}$ be the softmax output of model m for image i . The ensemble prediction is $\hat{y}_i = \arg \max_c \sum_{m=1}^M w_m p_{i,c}^{(m)}$, with three weighting schemes evaluated: **mean** ($w_m = 1/M$), **weighted** ($w_m \propto$ model m 's validation F1), and **adaptive** ($w_m \propto$ model m 's confidence on image i). The ensemble is computed entirely from the saved per-image probability files, requiring no GPU or model reloading.

G. Domain-Adaptation Fine-Tuning

To adapt a model to a new visual domain with minimal data, the backbone is kept frozen and only the classifier head is re-trained on a small set of target-domain images. This is evaluated with 5-fold cross-validation (30 epochs, $\text{lr}_{\text{head}} = 5 \times 10^{-4}$, batch size 16), reporting the mean macro F1 before and after adaptation.

H. Evaluation Metrics

Given the potential class imbalance in food datasets, accuracy alone is insufficient. Let TP_c , FP_c , FN_c denote true positives, false positives, and false negatives for class c , and C the number of classes. We report:

$$\overline{F1} = \frac{1}{C} \sum_{c=1}^C \frac{2TP_c}{2TP_c + FP_c + FN_c} \quad (2)$$

$$\text{MCC} = \frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}} \quad (3)$$

Additionally, we report Area Under the ROC Curve (AUC-ROC) and per-class F1, precision, and recall. For food safety applications, recall on the *rotten* class is particularly critical: a false negative (rotten food classified as fresh) poses a direct health risk.

IV. EXPERIMENTS

A. Phase 1: ResNet-18 Baseline

Table III reports the best macro F1-score achieved by ResNet-18 for each dataset and condition (state mode, 60 epochs). Condition C4 achieves the best result in four of six datasets; C2 wins on the two datasets where the number of training samples is smallest (KFQ: 287 train samples; MFV: 8,123 train samples but three-class complexity).

TABLE III: Best macro F1-score (%) — ResNet-18, state mode. Bold: best condition per dataset.

Dataset	C1	C2	C3	C4	Best
KFR	97.6	99.8	99.4	99.8	C2/C4
KFS	95.0	98.9	98.0	99.0	C4
MLM	96.2	98.0	96.2	98.5	C4
MFR	94.3	95.5	95.2	96.4	C4
MFV	78.3	91.0	86.7	90.6	C2
KFQ	86.1	91.6	86.0	88.9	C2

B. Phase 2: Backbone Comparison

Table IV compares the best macro F1-score per dataset across all four backbone architectures (state mode), with the winning condition shown in parentheses. EfficientNet-B0 achieves the best result on four of six datasets; MobileNetV3-Small — despite being the smallest backbone (2.5M parameters) — wins on the hardest three-class dataset (MFV) and ties for the lead on KFR and KFQ; ResNet-18 ties on KFQ, the smallest dataset. A clear pattern emerges in the *condition* column: every backbone except ResNet-18 attains its best result under C3 (frozen backbone + projection head), whereas ResNet-18 prefers C4 (partial fine-tuning).

TABLE IV: Best macro F1-score (%) by backbone, state mode. Winning condition in parentheses. Bold: overall best per dataset.

Dataset	ResNet-18	MobileNetV3	EffNet-B0	EffNet-B2
KFR	99.8 (C4)	100.0 (C3)	100.0 (C3)	99.9 (C3)
KFS	99.0 (C4)	99.2 (C3)	99.4 (C3)	99.3 (C3)
MLM	98.5 (C4)	98.2 (C3)	98.7 (C3)	98.0 (C3)
MFR	96.4 (C4)	96.7 (C3)	97.0 (C3)	96.4 (C3)
MFV	91.0 (C2)	92.4 (C3)	91.6 (C3)	92.3 (C3)
KFQ	91.6 (C2)	91.6 (C1)	86.1 (C4)	88.9 (C1)

C. Fruit-State Classification and Benchmark

To compare against published work, the two three-fruit datasets with existing baselines — KFR (apple/banana/orange) and MFR (peach/pomegranate/strawberry) — are trained in fruit_state mode (6 classes) with the full 4×4 grid of backbones and conditions. Tables V and VI report the macro F1 of every combination. The C3 dominance observed in state mode is even stronger here: C3 is the best condition for every backbone except ResNet-18 on both datasets, and EfficientNet-B0-C3 and MobileNetV3-C3 both reach a perfect 100% on KFR.

TABLE V: KFR fruit_state macro F1 (%). Bold: best per backbone.

Backbone	C1	C2	C3	C4
ResNet-18	97.78	99.46	99.40	99.65
MobileNetV3-S	98.71	97.25	100.0	99.20
EfficientNet-B0	99.70	98.75	100.0	99.46
EfficientNet-B2	99.53	98.46	99.97	99.36

Table VII places these results against the published baselines using accuracy, the metric all three papers report. On both datasets our models exceed the prior state of the art — by +0.35

TABLE VI: MFR fruit_state macro F1 (%). Bold: best per backbone.

Backbone	C1	C2	C3	C4
ResNet-18	90.98	92.67	94.09	93.35
MobileNetV3-S	93.13	89.48	95.25	93.84
EfficientNet-B0	93.36	84.05	95.72	88.49
EfficientNet-B2	92.82	82.08	93.98	84.71

points on KFR (ensemble vs. the best published MobileNetV2) and by +1.68 points on MFR (ensemble vs. the published ResNet50).

TABLE VII: Benchmark against published work on the same datasets (metric: accuracy, %). Bold: best per dataset.

Source / Model	Split	Acc.
<i>KFR — apple/banana/orange (6 classes)</i>		
Palakodati et al. [11], CNN	60-10-30	97.82
Chakraborty et al. [12], MobileNetV2	80-20	99.61
Ours — best single (EB0/MN3-C3)	80-20	100.0
Ours — ensemble (16 models)	80-20	99.96
<i>MFR — peach/pomegranate/strawberry (6 classes)</i>		
Sharma & Kumar [13], ResNet50	70-15-15	95.00
Ours — best single (EB0-C3)	80-20	95.77
Ours — ensemble (16 models)	80-20	96.68

D. Model Ensemble

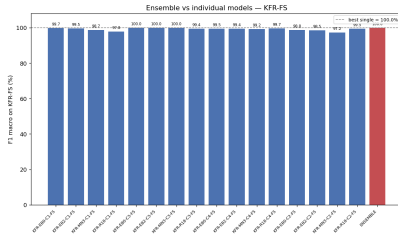
The sixteen fruit_state models per dataset are combined as described in Section III-F. Table VIII reports the outcome, which is almost invariant to the weighting scheme. The result is informative in both directions. On KFR, where the best single model is already perfect, the ensemble cannot improve and dips marginally (−0.03 pts): averaging a saturated model with weaker members only adds noise. On MFR, where headroom remains, the ensemble adds +0.85 pts of F1 (+0.96 pts of accuracy over the best single model), confirming that the four distinct backbone families produce complementary errors when the task is not already solved. Figure 1 shows the per-model F1 bars and the ensemble confusion matrix for both datasets.

TABLE VIII: Ensemble of 16 models (fruit_state). $\Delta F1$ is relative to the best single model.

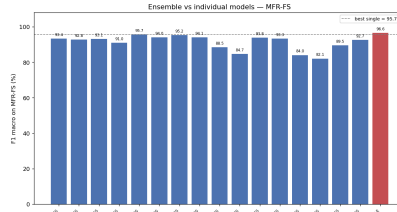
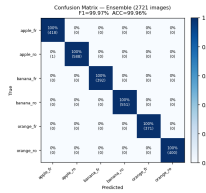
Data	Method	Best sgl.	Ens. F1	Ens. Acc	$\Delta F1$	MCC
KFR	mean	100.0	99.97	99.96	−0.03	0.9996
KFR	weighted	100.0	99.97	99.96	−0.03	0.9996
KFR	adaptive	100.0	99.97	99.96	−0.03	0.9996
MFR	mean	95.72	96.56	96.68	+0.84	0.9601
MFR	weighted	95.72	96.56	96.68	+0.84	0.9601
MFR	adaptive	95.72	96.57	96.68	+0.85	0.9601

E. Domain Shift and Per-Fruit Analysis

To measure generalization to real-world conditions, the state-mode models are evaluated on *own_dataset*: 219 mobile-camera photos of two fruits — 121 strawberry (64 fresh, 57 rotten) and 98 banana (46 fresh, 52 rotten) — used only for evaluation. Across all 80 evaluated models the mean F1 drop from training



(a) KFR — essentially perfect diagonal.



(b) MFR — residual confusion on rotten pomegranate/strawberry.

Fig. 1: Per-model macro F1 (left bars) and ensemble confusion matrix (right, percentages and counts) for the 16-model mean-softmax ensemble on each benchmark dataset.

to this set is 21.8 points (range -1.6 to 51.5). Crucially, the drop is not uniform across fruits but is governed by training exposure (Table IX): models whose source dataset *included* banana average 71.4% F1 on banana, whereas models that *never saw* banana collapse to 42.0% — barely above chance — while strawberry F1 is essentially unaffected. The worst performers in Table X are precisely the Lemon-Varieties (MLM) models, which never saw either evaluation fruit and fall below 55% despite $>92\%$ training F1.

TABLE IX: Per-fruit F1 on *own_dataset* (219 photos, state mode), averaged over the 80 models, grouped by whether banana appeared in training.

Model group	# models	Banana F1	Strawberry F1
Saw banana in training	48	71.4	80.5
Did not see banana	32	42.0	79.9

TABLE X: Cross-dataset evaluation on *own_dataset* (219 photos, state mode): top and bottom models by evaluation F1. Drop = $F1_{\text{train}} - F1_{\text{eval}}$.

Model ID	$F1_{\text{train}}$	$F1_{\text{eval}}$	Drop	Saw banana
KFS-EB0-C3	99.4	86.7	12.7	yes
KFQ-EB0-C1	84.7	86.3	-1.6	yes
KFS-R18-C3	98.0	84.4	13.6	yes
KFQ-R18-C3	86.0	83.5	2.5	yes
MLM-EB2-C2	92.6	54.6	38.0	no
MLM-EB2-C4	94.1	52.0	42.1	no
MLM-MN3-C4	92.8	41.6	51.3	no
MLM-MN3-C2	91.0	39.6	51.5	no

a) Domain-adaptation fine-tuning.: The domain gap can be closed cheaply. Starting from the EfficientNet-B0 head trained on Fruits Quality (KFQ-EB0-C1) and fine-tuning only the head on *own_dataset* with the backbone frozen (Section III-G), macro F1 rises from 86.3% to **94.5%** (+8.2 points) under 5-fold cross-validation, with the safety-critical rotten-class recall improving in step. Because only the lightweight head is updated, this adaptation takes seconds and needs no backbone storage — a practical recipe for deployment: pre-train on abundant public images, then fine-tune the head on a small set of in-domain photos.

V. RESULTS AND DISCUSSION

Finding 1: C3 is optimal for MobileNet/EfficientNet; C4 for ResNet-18. Across both state and fruit_state modes, MobileNetV3-Small and both EfficientNet variants achieve their best performance under C3 (frozen backbone + projection head), while ResNet-18 consistently benefits from partial fine-tuning (C4). We attribute this to the richer pre-trained features of the MobileNet/EfficientNet families: their 576–1408-dimensional outputs already capture discriminative cues for freshness without backbone adaptation, whereas ResNet-18’s 512-dimensional output benefits from specialized high-level feature adaptation. Fine-tuning these backbones with the available data sizes appears to introduce overfitting rather than beneficial specialization.

Finding 2: Dataset size determines the winning backbone. Below approximately 500 training samples, the compact backbones (ResNet-18 and MobileNetV3-Small, both 91.6% on KFR) outperform EfficientNet-B0 by up to 5.5 percentage points (86.1%). Above 1,500 training samples, EfficientNet-B0 consistently matches or surpasses the other backbones, winning four of six datasets. This is consistent with the larger head of EfficientNet requiring more data to generalize.

Finding 3: Our models exceed the published state of the art. On both datasets with published baselines, our results surpass prior work (Table VII). On KFR our best single model (EfficientNet-B0/MobileNetV3 under C3) reaches a perfect 100% accuracy and the ensemble 99.96%, above the 99.61% reported with MobileNetV2 [12]. On MFR the ensemble reaches 96.68%, above the 95.0% reported with ResNet50 [13]. The perfect KFR result ($MCC = 1.0$, $AUC-ROC = 1.0$) indicates that the fresh/rotten distinction in this dataset is highly discriminative and fully captured by frozen pre-trained features.

Finding 4: Ensembling helps only where headroom remains. Combining the 16 fruit_state models yields opposite outcomes on the two datasets (Table VIII). On KFR, where the best single model is already perfect, the ensemble marginally dips (-0.03 pts) because averaging a saturated model with weaker members can only add noise. On MFR, where the best single model reaches 95.72%, the ensemble adds +0.85 pts of F1. The effect is invariant to the weighting scheme, confirming that the gain stems from genuine error diversity across the four backbone families rather than from any particular weighting

heuristic.

Finding 5: Formalin remains a fundamental challenge. FruitVision is the only dataset where no backbone exceeds 92.4% F1-score, despite having 10,154 images. The formalin class is chemically treated to appear visually similar to fresh samples, creating a subtle shift difficult to detect from RGB images alone. MobileNetV3-Small achieves the best result (92.4%), narrowly ahead of EfficientNet-B2 (92.3%). This is directly relevant to detecting subtle spoilage in fermented foods.

Finding 6: Generalization tracks per-fruit training exposure. On the 219-photo real-world set, a model’s accuracy on a given fruit is governed by whether that fruit appeared in training (Table IX): banana F1 averages 71.4% for models that saw banana versus 42.0% for those that did not, while strawberry F1 is essentially unchanged ($\approx 80\%$ either way). This explains the bottom of Table X: the worst models are all Lemon-Varieties models, which never saw either evaluation fruit and so fail catastrophically despite $>92\%$ training F1. High in-domain accuracy therefore does not imply good generalization to unseen fruits or visual domains.

Finding 7: Lightweight head fine-tuning closes the domain gap. Freezing the backbone and re-training only the classifier head on a small set of in-domain photos raises macro F1 from 86.3% to 94.5% (+8.2 points) under 5-fold cross-validation. Because only the head is updated, the adaptation is fast and storage-free, offering a practical deployment recipe: pre-train on abundant public images, then adapt the head to the target domain.

VI. CONCLUSION

This paper presented a systematic comparison of transfer learning strategies for fruit and vegetable freshness classification. Four backbone architectures (ResNet-18, MobileNetV3-Small, EfficientNet-B0, EfficientNet-B2) and four training conditions were evaluated across six benchmark datasets totalling 55,040 images, in both binary and six-class formulations, and combined into per-dataset ensembles. Generalization was measured against 219 real-world photos of two fruits.

The key findings are: (1) on the two datasets with published baselines, our models exceed the state of the art — a perfect 100% accuracy (best single) and 99.96% (ensemble) on the apple/banana/orange benchmark, and 96.68% (ensemble) on the peach/pomegranate/strawberry benchmark; (2) the optimal training condition differs by backbone family — C4 for ResNet-18, C3 (frozen backbone + projection head) for the MobileNet and EfficientNet families — indicating that richer pre-trained features reduce the need for backbone fine-tuning; (3) ensembling the four backbone families helps only where headroom remains, adding +0.85 F1 points on the harder dataset but slightly reducing the already-saturated one; (4) real-world generalization is governed by per-fruit training exposure, with banana F1 falling from 71% to 42% when the fruit is unseen; (5) a lightweight, backbone-frozen head fine-tuning step raises in-domain macro F1 from 86.3% to 94.5%, providing a practical domain-adaptation recipe.

Future work will extend this pipeline to Korean food categories (kimchi, rice cake, fermented side dishes), where the visual similarity between fresh and fermented states is directly analogous to the formalin challenge identified in this study, applying the same public-pretrain then head-adapt recipe to bridge the domain gap.

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